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Advanced Tic Tac Toe Game

Qt Test document

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Reading Report: Qt Test Framework

Submitted as part of the Advanced Tic Tac Toe Project

Introduction

This document describes the testing process for the Tic Tac Toe game developed using Qt. The application includes GUI-based interaction for user registration, login and player-vs-computer game modes, and leaderboard tracking.

Testing is conducted to ensure:

- Functional correctness of game logic and GUI behavior
- File-based persistence (e.g., login data and match history)
- Compatibility of GUI elements with expected user flows

Overview of Qt Test (QtTest Module)

Qt Test is a module provided by the Qt framework to support automated unit testing of both logic and GUI applications. It enables developers to write test cases for individual components, functions, and even simulate user interaction with widgets.

Key features:

- Based on QTestLib, the built-in Qt unit test framework
- Supports QVERIFY, QCOMPARE, and other macros for validation

File Handling Testing

- Tested users.txt, history.txt, and <username>_history.txt for:
- Append-only behavior
- Data format correctness (tab-separated or colon-separated)
- File creation if missing
- Used QDir::currentPath() for debugging file access paths.

Tested Functional Areas

During the development and testing phase of the project, several core functionalities were carefully verified through extensive manual testing and functional validation. These tests aimed to ensure both

correct behavior and user experience quality across the most essential features of the application.

1. User Registration

The registration system was tested to confirm that:

- New users can be successfully created and stored in the system.
- Registration data is consistently written to the appropriate file (users.txt) with the correct format.

This ensures the system maintains a reliable and unique user base.

2. User Login

Login functionality was tested for both valid and invalid credentials:

- Verified that users can log in successfully when using correct usernames and passwords.
- Attempted logins with incorrect credentials were properly denied, with appropriate error feedback.
- The system reads and compares data from users.txt accurately, using proper string parsing and logic.

These tests confirmed that user authentication works as intended and prevents unauthorized access.

3. Player vs Computer Game Mode

This game mode was a major focus of the testing process, covering the following:

- The player is allowed to choose their preferred symbol (X or O), and the game responds correctly based on that choice.
- AI moves were checked across different difficulty levels (Easy, Medium, Hard) to confirm they are valid, timely, and responsive.
- Game outcomes (win, lose, draw) are detected accurately.

All logic related to gameplay, turn switching, and AI behavior was verified to be working seamlessly.

4. Leaderboard Tracking

Leaderboard functionality was tested to ensure proper tracking of match results:

- After every completed game, the results are automatically logged and appended to score.txt.
- Each player's win, loss, or draw count is recorded and updated accurately.

- The leaderboard display retrieves data correctly from the file and shows meaningful statistics to the user.

This guarantees that player performance is persistently and correctly stored across sessions.

Conclusion

The testing process for the Tic Tac Toe project ensured that all key functionalities—specifically user registration, login, player-versus-computer gameplay, and leaderboard tracking—operate reliably and as intended. Through extensive manual and functional testing, the application was verified to handle standard interactions correctly.

The GUI responded appropriately to user inputs, symbol selection, and turn-based gameplay, including correct handling of AI logic and game outcomes. All associated file operations for saving user data, match history, and leaderboard results were also confirmed to work without errors.